Converting Between Gaussian and Lognormal Families

Two strategies — a bridge to Girsanov

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1 The program

Our ultimate goal is to understand how geometric Brownian motion (GBM) is converted into a martingale under the risk-neutral measure — the core of Black–Scholes. That conversion is an instance of Girsanov's theorem, which operates on infinite-dimensional path space and is technically demanding.

The strategy here is to build intuition through finite-dimensional analogues first, working through two families of distributions and two conversion strategies before lifting to processes. Each step will reappear, in more complex form, in the Girsanov argument.

Recall the two strategies (from the companion notes):

Strategy	**Action**	**Keeps fixed**
Way 1	Change the measure on source space Ω	the map $f : \Omega \to \mathbb{R}$
Way 2	Change the map $f \to g$	the measure on Ω

The simplest setting: $\Omega = \mathbb{R}$. When $\Omega = \mathbb{R}$ and $f = \mathrm{id}$, Way 1 means replacing $\mu_1 = \mathcal{N}(\mu_1, \sigma_1^2)$ directly with a reweighted version on $\mathbb{R}$. On $\Omega = \mathbb{R}$ we have explicit control over the measure.

We work through four cases in order:

Section	**Source**	**Target**
§2	$\mathcal{N}(\mu_1, \sigma_1^2)$	$\mathcal{N}(\mu_2, \sigma_2^2)$ via Way 2 (affine map)
§3	$\mathcal{N}(\mu_1, \sigma_1^2)$	$\mathcal{N}(\mu_2, \sigma_2^2)$ via Way 1 (reweight)
§4	$\mathrm{LogN}(\mu_1, \sigma_1^2)$	$\mathrm{LogN}(\mu_2, \sigma_2^2)$ via Way 2 (power map)
§5	$\mathrm{LogN}(\mu_1, \sigma_1^2)$	$\mathrm{LogN}(\mu_2, \sigma_2^2)$ via Way 1 (reweight)

2 Normal family, Way 2: the affine map

2.1 The map

Let $\Omega = \mathbb{R}$, $\mu_1 = \mathcal{N}(\mu_1, \sigma_1^2)$ on Ω, and $f_1 = \mathrm{id}$, so the induced measure on $B = \mathbb{R}$ is $\nu_1 = \mathcal{N}(\mu_1, \sigma_1^2)$.

We want $\nu_2 = \mathcal{N}(\mu_2, \sigma_2^2)$ on B. As discussed in the companion notes, this is achieved by a map $g : B \to B$ on the target, with $f_2 = g \circ f_1 = g$.

The standard normal $Z \sim \mathcal{N}(0, 1)$ satisfies $\mu_1 + \sigma_1 Z \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $\mu_2 + \sigma_2 Z \sim \mathcal{N}(\mu_2, \sigma_2^2)$. So we need a map sending $\mu_1 + \sigma_1 z \mapsto \mu_2 + \sigma_2 z$. Solving: $z = (y - \mu_1)/\sigma_1$, so the map is

$$g(y) = \mu_2 + \frac{\sigma_2}{\sigma_1}(y - \mu_1),$$

an affine map: scale by σ_2/σ_1 around μ_1, then shift to μ_2.

2.2 Verification by change of variables

g is strictly increasing with $g'(y) = \sigma_2/\sigma_1$ and $g^{-1}(y) = \mu_1 + \frac{\sigma_1}{\sigma_2}(y - \mu_2)$. The pushforward density (Form A):

$$f_{\nu_2}(y) = f_{\nu_1}(g^{-1}(y)) \cdot \frac{1}{|g'(g^{-1}(y))|} = \frac{1}{\sigma_1}\phi\left(\frac{g^{-1}(y) - \mu_1}{\sigma_1}\right) \cdot \frac{\sigma_1}{\sigma_2} = \frac{1}{\sigma_2}\phi\left(\frac{y - \mu_2}{\sigma_2}\right),$$

which is the $\mathcal{N}(\mu_2, \sigma_2^2)$ density. ✓

2.3 Special cases

Case	**Map** $g(y)$	**Effect**
$\sigma_1 = \sigma_2,\ \mu_1 \neq \mu_2$	$y + (\mu_2 - \mu_1)$	pure shift
$\mu_1 = \mu_2,\ \sigma_1 \neq \sigma_2$	$\mu_1 + \dfrac{\sigma_2}{\sigma_1}(y - \mu_1)$	scale around mean
$\mu_1 = 0,\ \sigma_1 = 1$ (standard)	$\mu_2 + \sigma_2 y$	the standard parameterisation

2.4 The affine group and its inversion operator

The normal-to-normal map is most cleanly understood through the group of affine functions on $\mathbb{R}$.

The affine family. Define

$$\mathrm{Af} = \big\{ f : \mathbb{R} \to \mathbb{R} \mid f \text{ is affine} \big\} = \big\{ \mathrm{af}_{a,b}(x) = a + bx \ \big|\ a, b \in \mathbb{R} \big\}.$$

Af is closed under composition: $\mathrm{af}_{a_2,b_2} \circ \mathrm{af}_{a_1,b_1} = \mathrm{af}_{a_2+b_2a_1,\, b_2b_1}$. A map $\mathrm{af}_{a,b}$ is invertible if and only if $b \neq 0$, in which case

$$y = a + bx \implies x = \frac{y - a}{b} = -\frac{a}{b} + \frac{1}{b}\,y = \mathrm{af}_{-a/b,\, 1/b}(y).$$

The coefficient operator for inversion. The passage from a function to its inverse corresponds to a map on the parameter pairs (a, b). Define the *coefficient operator*

$$\boxed{C :\ (a, b) \mapsto \left(-\frac{a}{b},\ \frac{1}{b}\right)}, \qquad C : \mathbb{R} \times \mathbb{R} \setminus \{0\} \ \longrightarrow\ \mathbb{R} \times \mathbb{R} \setminus \{0\}.$$

Then $(\mathrm{af}_{a,b})^{-1} = \mathrm{af}_{C(a,b)}$.
Verification. C is an involution: $C(C(a, b)) = C(-a/b, 1/b) = (-(-a/b)/(1/b),\, b) = (a, b)$. This reflects the fact that inverting twice returns the original map. ✓

2.5 How C acts on lines in parameter space

Every affine map $\mathrm{af}_{a,b}$ corresponds to a point (a, b) in the parameter plane $\mathbb{R} \times (\mathbb{R} \setminus \{0\})$. Families of affine maps that share a structural constraint correspond to *curves* in this plane. The operator C acts on the whole plane, and it is illuminating to track what it does to the most natural curves: lines.

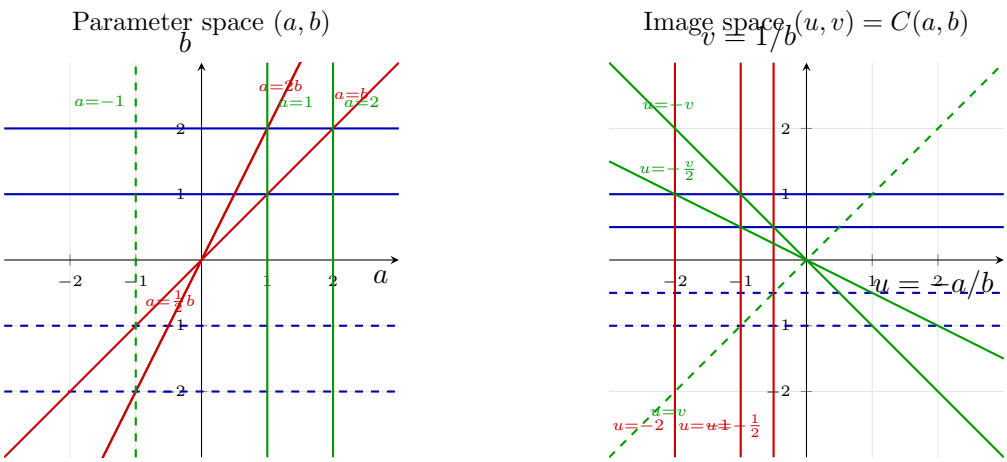
Three canonical line families and their images under C.

Let $(u, v) = C(a, b) = (-a/b,\ 1/b)$. Three families of lines in (a, b)-space and their C-images in (u, v)-space:

Family in (a, b)	**Condition**	**Image under** C	**Type**
Horizontal: $b = c$	b fixed, a free	$v = 1/c$ (horizontal line)	horiz. $\to$ horiz.
Ray through origin: $a = \lambda b$	slope λ, $a/b = \lambda$	$u = -\lambda$ (vertical line)	ray $\to$ vertical
Vertical: $a = c$	a fixed, b free	$u = -cv$ (ray through 0)	vertical $\to$ ray

Note the duality: C swaps "vertical lines" with "rays through the origin", and maps horizontal lines to horizontal lines (at the reciprocal height). This is a consequence of the involution property $C^2 = \mathrm{id}$.

The two plots below show the three families before and after applying C. Each coloured line in the left panel maps to the correspondingly coloured curve/line in the right panel.



Left: three families of lines in the (a, b) parameter plane. Right: their images under $C : (a, b) \mapsto (-a/b,\ 1/b)$. Blue (horizontal) $\mapsto$ horizontal at reciprocal height; Red (rays through 0) $\mapsto$ vertical lines; Green (vertical) $\mapsto$ rays through 0. The colour-matched duality reflects $C^2 = \mathrm{id}$.

Geometric interpretation. The three families form a complete picture of the affine parameter plane:

- **Horizontal lines** $b = c$: all maps with the *same slope*. C maps them to horizontal lines at height $1/c$, i.e. slope inverts. This corresponds to: inverting an affine map replaces slope b by $1/b$.
- **Rays through the origin** $a = \lambda b$: all maps whose *fixed point* is at $x = -a/(b-1)$; the ratio $a/b = \lambda$ is preserved along the ray. C sends each ray to the vertical line $u = -\lambda$, collecting all maps with the same ratio into a single fibre.
- **Vertical lines** $a = c$: all maps with the same *intercept*. C maps them to rays through the origin $u = -cv$, i.e. $a/b = -c \cdot (1/b)$, so the ratio a/b is now proportional to the new slope $1/b$.

Application to standardisation. The standardisation maps $\mathrm{af}_i = \mathrm{af}_{-\mu_i/\sigma_i,\,1/\sigma_i}$ lie on the ray $a = -\mu_i \cdot b$ (passing through the origin with slope $-\mu_i$ in (a,b)-space) intersected with the horizontal line $b = 1/\sigma_i$. Their images under C are the de-standardisation maps $\mathrm{af}_i^{-1} = \mathrm{af}_{\mu_i,\sigma_i}$, which lie on the vertical line $a = \mu_i$ (in (u,v)-space: $u = \mu_i$... wait — actually $C(-\mu_i/\sigma_i, 1/\sigma_i) = (\mu_i, \sigma_i)$, a single point, not a line). The graphical picture applies to *one-parameter families*; for isolated points, C simply maps point to point.

2.6 Standardisation maps and the factored route to g

Every $\mathcal{N}(\mu, \sigma^2)$ distribution is related to the standard normal $\mathcal{N}(0,1)$ by an affine map. Define the *standardisation map*

$$\mathrm{af}_i \;:=\; \mathrm{af}_{-\mu_i/\sigma_i,\,1/\sigma_i}, \qquad \mathrm{af}_i(x) = \frac{x-\mu_i}{\sigma_i},$$

so that if $X \sim \mathcal{N}(\mu_i, \sigma_i^2)$ then $\mathrm{af}_i(X) \sim \mathcal{N}(0,1)$.

The inverse is

$$\mathrm{af}_i^{-1} = \mathrm{af}_{C(-\mu_i/\sigma_i,\,1/\sigma_i)} = \mathrm{af}_{\mu_i,\sigma_i}, \qquad \mathrm{af}_i^{-1}(z) = \mu_i + \sigma_i z.$$

The Normal-to-Normal map $g : \mathcal{N}(\mu_1, \sigma_1^2) \to \mathcal{N}(\mu_2, \sigma_2^2)$ factors through $\mathcal{N}(0,1)$:

$$\begin{tikzcd}
\mathcal{N}(\mu_1,\sigma_1^2) \arrow[r, "\mathrm{af}_1"] \arrow[d, dashed, "g = \mathrm{af}_2^{-1} \circ \mathrm{af}_1"'] & \mathcal{N}(0,1) \arrow[dl, "\mathrm{af}_2^{-1}"] \\
\mathcal{N}(\mu_2,\sigma_2^2) &
\end{tikzcd}$$

Deriving g from the composition.

$$g(y) = \big(\mathrm{af}_2^{-1} \circ \mathrm{af}_1\big)(y) = \mathrm{af}_2^{-1}\left(\frac{y-\mu_1}{\sigma_1}\right) = \mu_2 + \sigma_2 \cdot \frac{y-\mu_1}{\sigma_1} = \mu_2 + \frac{\sigma_2}{\sigma_1}(y-\mu_1).$$

This agrees exactly with the map derived in §2. ✓

2.7 Inversion of the composition and the inverse map g^{-1}

Inverting the composition. Since $g = \mathrm{af}_2^{-1} \circ \mathrm{af}_1$, its inverse is obtained by reversing the order and inverting each factor:

$$g^{-1} = \big(\mathrm{af}_2^{-1} \circ \mathrm{af}_1\big)^{-1} = \mathrm{af}_1^{-1} \circ \mathrm{af}_2.$$

Explicitly:

$$g^{-1}(x) = \mathrm{af}_1^{-1}\left(\frac{x-\mu_2}{\sigma_2}\right) = \mu_1 + \sigma_1 \cdot \frac{x-\mu_2}{\sigma_2} = \mu_1 + \frac{\sigma_1}{\sigma_2}(x-\mu_2).$$

This matches the formula for g^{-1} in §2 (the variable label $y \leftrightarrow x$ is cosmetic). ✓

Symmetry. The inversion formula has a transparent structure: swapping the index labels $1 \leftrightarrow 2$ in g gives g^{-1}:

$$g(y) = \mu_2 + \frac{\sigma_2}{\sigma_1}(y-\mu_1) \quad \xrightarrow{1\leftrightarrow 2} \quad g^{-1}(x) = \mu_1 + \frac{\sigma_1}{\sigma_2}(x-\mu_2).$$

This is a direct consequence of the group structure: C applied to the pair $(a,b) = (-\mu_2/\sigma_1 + \mu_1\sigma_2/\sigma_1^2,\ \sigma_2/\sigma_1)$... or more transparently, the composition route $\mathrm{af}_1^{-1} \circ \mathrm{af}_2$ automatically produces the swapped formula.

2.8 Homework: verification via the Jacobian formula

Exercise. Verify directly, using the Jacobian (change-of-variables) formula, that $g_*\mathcal{N}(\mu_1, \sigma_1^2) = \mathcal{N}(\mu_2, \sigma_2^2)$.

Solution. We apply Form A: if g is strictly increasing then

$$f_{\nu_2}(y) = f_{\nu_1}(g^{-1}(y)) \cdot |(g^{-1})'(y)|.$$

We have $g^{-1}(y) = \mu_1 + \frac{\sigma_1}{\sigma_2}(y - \mu_2)$, so $(g^{-1})'(y) = \sigma_1/\sigma_2$ (constant, positive). Substituting:

$$\begin{aligned} f_{\nu_2}(y) &= \frac{1}{\sigma_1}\,\phi\left(\frac{g^{-1}(y) - \mu_1}{\sigma_1}\right) \cdot \frac{\sigma_1}{\sigma_2} \\ &= \frac{1}{\sigma_2}\,\phi\left(\frac{\mu_1 + \frac{\sigma_1}{\sigma_2}(y - \mu_2) - \mu_1}{\sigma_1}\right) \\ &= \frac{1}{\sigma_2}\,\phi\left(\frac{y - \mu_2}{\sigma_2}\right), \end{aligned}$$

which is the $\mathcal{N}(\mu_2, \sigma_2^2)$ density. ✓

Remark on the two routes. The computation in §2 used the formula with g and g' directly (Form A in the g direction); the exercise above uses g^{-1} and $(g^{-1})'$. Both are instances of the same change-of-variables theorem and give identical results, as expected.

2.9 Affine pushforward from first principles

Remark 1 (Derivation from first principles)**.** *The following two sub-cases derive the pushed-forward density both from the definition of pushforward measure and by the Jacobian formula, then reconciling the two routes. The purely linear case ($a = 0$) is treated separately to isolate the role of the slope b.*

We fix $\lambda =$ Lebesgue measure throughout. An important preliminary: if μ has a Lebesgue density f_μ and $h : \mathbb{R} \to \mathbb{R}$ is a Lebesgue-measure-preserving-class map (e.g. strictly monotone), then the pushforward $\nu = h_*\mu$ also has a Lebesgue density f_ν. This fails for general reference measures (e.g. a shift of a measure supported on $[c, d]$ may leave the support of the reference), but holds automatically for Lebesgue measure on $\mathbb{R}$.

Case A: pure shift $h(x) = x + a$

Let $h(x) = x + a$, $a \in \mathbb{R}$, so h maps the interval $[c, d]$ to $[c + a, d + a]$.

From the definition. The pushforward condition $\nu(B) = \mu(h^{-1}(B))$ applied to $B = [c+a, d+a]$ gives

$$\nu([c + a, d + a]) = \mu([c, d]) \implies \int_{c+a}^{d+a} f_\nu(y)\ \mathrm{d}y = \int_c^d f_\mu(x)\ \mathrm{d}x.$$

Change variable on the left: $y = x + a$, $\mathrm{d}y = \mathrm{d}x$, limits $c \to c + a$, $d \to d + a$:

$$\int_{c+a}^{d+a} f_\nu(y)\ \mathrm{d}y = \int_{c+a}^{d+a} f_\mu(y - a)\ \mathrm{d}y.$$

Since this holds for every interval $[c, d]$, equating integrands gives

$$\boxed{f_\nu(y) = f_\mu(y - a).}$$

Confirmation by the Jacobian formula. Here $h^{-1}(y) = y - a$ and $(h^{-1})'(y) = 1$, so the Jacobian is $|J| = |(h^{-1})'| = 1$. The pushforward density formula gives

$$f_\nu(y) = |J| \cdot f_\mu(h^{-1}(y)) = 1 \cdot f_\mu(y-a) = f_\mu(y-a). \checkmark$$

Normal example. Let $\mu = \mathcal{N}(\mu_1, \sigma^2)$ and $a = \mu_2 - \mu_1$. Then

$$f_\nu(y) = f_\mu(y-a) = \frac{1}{\sigma}\phi\left(\frac{(y-a)-\mu_1}{\sigma}\right) = \frac{1}{\sigma}\phi\left(\frac{y-\mu_2}{\sigma}\right),$$

so $\nu = \mathcal{N}(\mu_2, \sigma^2)$. A pure shift of the density by $a = \mu_2 - \mu_1$ produces the target normal with the same variance. ✓

Case B: purely linear map $h(x) = bx$, $b > 0$

Let $h(x) = bx$, so h maps $[c, d]$ to $[bc, bd]$.

From the definition.

$$\mu([c,d]) = \nu([bc,bd]) \implies \int_c^d f_\mu(x)\ \mathrm{d}x = \int_{bc}^{bd} f_\nu(y)\ \mathrm{d}y.$$

Change variable on the left: $y = bx$, $x = y/b$, $\mathrm{d}x = \frac{1}{b}\,\mathrm{d}y$, limits $c \to bc$, $d \to bd$:

$$\int_c^d f_\mu(x)\ \mathrm{d}x = \int_{bc}^{bd} f_\mu\left(\frac{y}{b}\right)\frac{1}{b}\ \mathrm{d}y.$$

Equating integrands:

$$\boxed{f_\nu(y) = \frac{1}{b}\, f_\mu\left(\frac{y}{b}\right).}$$

Confirmation by the Jacobian formula. $h^{-1}(y) = y/b$, $(h^{-1})'(y) = 1/b$, so $|J| = 1/b$.

$$f_\nu(y) = |J| \cdot f_\mu(h^{-1}(y)) = \frac{1}{b} \cdot f_\mu\left(\frac{y}{b}\right). \checkmark$$

Same result as from first principles.

Normal example. Let $\mu = \mathcal{N}(\mu_1, \sigma_1^2)$ and $b = \sigma_2/\sigma_1$. Then $h^{-1}(y) = y\sigma_1/\sigma_2$ and

$$\begin{aligned} f_\nu(y) &= \frac{1}{b} \cdot f_\mu\left(\frac{y}{b}\right) = \frac{\sigma_1}{\sigma_2} \cdot \frac{1}{\sigma_1}\phi\left(\frac{y\sigma_1/\sigma_2 - \mu_1}{\sigma_1}\right) \\ &= \frac{1}{\sigma_2}\,\phi\left(\frac{y/\sigma_2 - \mu_1/\sigma_1}{1}\right) = \frac{1}{\sigma_2}\,\phi\left(\frac{y - (\sigma_2/\sigma_1)\mu_1}{\sigma_2}\right), \end{aligned}$$

so $\nu = \mathcal{N}\big((\sigma_2/\sigma_1)\mu_1,\ \sigma_2^2\big)$.

Key observation. The purely linear map $x \mapsto bx$ shifts the mean as well as rescales the variance: it maps $\mathcal{N}(\mu_1, \sigma_1^2)$ to $\mathcal{N}(b\mu_1, b^2\sigma_1^2)$, not to $\mathcal{N}(\mu_2, \sigma_2^2)$ for an arbitrary μ_2. In the case $b = \sigma_2/\sigma_1$ this gives $\mathcal{N}\big(\frac{\sigma_2}{\sigma_1}\mu_1,\ \sigma_2^2\big)$, which equals $\mathcal{N}(\mu_2, \sigma_2^2)$ only when $\mu_2 = \frac{\sigma_2}{\sigma_1}\mu_1$, i.e. when the two means are already in the correct ratio. To hit an arbitrary target mean μ_2, one must follow the linear map with a shift of $\mu_2 - \frac{\sigma_2}{\sigma_1}\mu_1$ — which is precisely the constant term $a = \mu_2 - b\mu_1$ in the full affine map $g(x) = \mu_2 + \frac{\sigma_2}{\sigma_1}(x - \mu_1)$ of §2.

Summary: the full affine case $h(x) = a + bx$

Combining Cases A and B (or applying either formula directly with $h^{-1}(y) = (y-a)/b$, $(h^{-1})' = 1/b$):

$$f_\nu(y) = \frac{1}{|b|} f_\mu\left(\frac{y-a}{b}\right).$$

For $\mu = \mathcal{N}(\mu_1, \sigma_1^2)$, $a = \mu_2 - \frac{\sigma_2}{\sigma_1}\mu_1$, $b = \sigma_2/\sigma_1$:

$$f_\nu(y) = \frac{\sigma_1}{\sigma_2} \cdot \frac{1}{\sigma_1}\phi\left(\frac{(y-a)/\sigma_2 - \mu_1/\sigma_1}{1}\right) = \frac{1}{\sigma_2}\phi\left(\frac{y-\mu_2}{\sigma_2}\right),$$

so $\nu = \mathcal{N}(\mu_2, \sigma_2^2)$. ✓ This is the same result as §2 and §2.8, now derived a third time from the measure-theoretic definition.

2.10 Having a λ-density does not guarantee $\mathrm{d}\nu/\mathrm{d}\mu$ exists

The preliminary statement in §2.9 says: if $\mu \ll \lambda$ then $\nu = h_*\mu \ll \lambda$ as well, i.e. the pushforward inherits absolute continuity with respect to Lebesgue measure. It is tempting to conclude that $\nu \ll \mu$ also holds, i.e. that $\mathrm{d}\nu/\mathrm{d}\mu$ exists. **This is false in general.**

Having a λ-density is a statement about the relationship of each measure to Lebesgue measure separately:

$$\mu \ll \lambda \quad \text{and} \quad \nu \ll \lambda \quad \not\Longrightarrow \quad \nu \ll \mu.$$

The Radon–Nikodym derivative $\mathrm{d}\nu/\mathrm{d}\mu$ exists if and only if $\nu \ll \mu$, i.e. every μ-null set is also ν-null. Two measures can both be absolutely continuous with respect to Lebesgue measure while being mutually singular with respect to each other.

Counterexample: uniform measures on disjoint intervals.

Let

$$\mu = \mathrm{Uniform}[c_1, d_1], \qquad \nu = \mathrm{Uniform}[c_2, d_2],$$

where $[c_1, d_1]$ and $[c_2, d_2]$ are two intervals with *empty intersection.* Both measures have Lebesgue densities:

$$f_\mu(x) = \frac{\mathbf{1}_{[c_1,d_1]}(x)}{d_1 - c_1}, \qquad f_\nu(x) = \frac{\mathbf{1}_{[c_2,d_2]}(x)}{d_2 - c_2},$$

so $\mu \ll \lambda$ and $\nu \ll \lambda$. Nevertheless:

- $\mu([c_1, d_1]) = 1$ but $\nu([c_1, d_1]) = 0$ (since the intervals are disjoint). Hence the set $[c_1, d_1]$ has full μ-measure but zero ν-measure: ν is *not* absolutely continuous with respect to μ.
- Symmetrically, $\nu([c_2, d_2]) = 1$ but $\mu([c_2, d_2]) = 0$, so μ is not absolutely continuous with respect to ν either.

The two measures are **mutually singular**: $\mu \perp \nu$. One can write $\mathbb{R} = [c_1, d_1] \cup ([c_1, d_1])^c$ with $\mu([c_1, d_1]) = 1$ and $\nu([c_1, d_1]) = 0$, which is the definition of mutual singularity. Neither $\mathrm{d}\nu/\mathrm{d}\mu$ nor $\mathrm{d}\mu/\mathrm{d}\nu$ exists.

Why Gaussians are special. The reason $\mathrm{d}\nu/\mathrm{d}\mu$ *does* exist between two Gaussian measures $\mathcal{N}(\mu_1, \sigma_1^2)$ and $\mathcal{N}(\mu_2, \sigma_2^2)$ is not merely that both have Lebesgue densities. The key is that both densities are *strictly positive on all of* $\mathbb{R}$: for any Borel set A,

$$\mu(A) = 0 \iff \lambda(A) = 0 \iff \nu(A) = 0.$$

Hence μ and ν share exactly the same null sets (the Lebesgue null sets), giving mutual absolute continuity $\mu \sim \nu$. The formula $\mathrm{d}\nu/\mathrm{d}\mu = f_\nu/f_\mu$ is then well-defined everywhere that $f_\mu > 0$, which is all of $\mathbb{R}$.

Contrast with the uniform case: f_μ vanishes outside $[c_1, d_1]$, so the ratio f_ν/f_μ is not defined on $[c_2, d_2]$ (where $f_\nu > 0$ but $f_\mu = 0$), making a Radon–Nikodym derivative impossible.

Structural summary.

Condition	**Guarantees**	**Example**
$\mu \ll \lambda$	f_μ exists	any measure with density
$\mu \ll \lambda$ and $\nu = h_*\mu$	f_ν exists	affine pushforward
$\mu \ll \lambda$ and $\nu \ll \lambda$	f_μ, f_ν both exist	two uniforms on $[0,1]$, $[2,3]$
(above) + disjoint supports	$\mu \perp \nu$, no RN derivative	uniforms on $[0,1]$, $[2,3]$
$f_\mu > 0$ λ-a.e. and $f_\nu > 0$ λ-a.e.	$\mu \sim \nu$, RN exists both ways	Gaussians, exponentials

3 Normal family, Way 1: reweighting the measure

3.1 Absolute continuity

Keep the map $f_1 = \mathrm{id}$ fixed. We ask: can we replace $\mu_1 = \mathcal{N}(\mu_1, \sigma_1^2)$ with $\mu_2 = \mathcal{N}(\mu_2, \sigma_2^2)$ by reweighting?

Both measures have Lebesgue densities that are strictly positive everywhere on $\mathbb{R}$, so they share the same null sets (the Lebesgue null sets). Therefore $\mu_1 \sim \mu_2$ (mutual absolute continuity), and both $\mathrm{d}\mu_2/\mathrm{d}\mu_1$ and $\mathrm{d}\mu_1/\mathrm{d}\mu_2$ exist.

3.2 The Radon–Nikodym derivative

$$p(x) = \frac{\mathrm{d}\mathcal{N}(\mu_2, \sigma_2^2)}{\mathrm{d}\mathcal{N}(\mu_1, \sigma_1^2)}(x) = \frac{f_{\mu_2}(x)}{f_{\mu_1}(x)} = \frac{\sigma_1}{\sigma_2} \exp\left(\frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(x-\mu_2)^2}{2\sigma_2^2} \right).$$

The exponent is a quadratic in x. Expanding:

$$\frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(x-\mu_2)^2}{2\sigma_2^2} = \frac{1}{2}\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right) x^2 + \left(\frac{\mu_2}{\sigma_2^2} - \frac{\mu_1}{\sigma_1^2}\right) x + \left(\frac{\mu_1^2}{2\sigma_1^2} - \frac{\mu_2^2}{2\sigma_2^2}\right). \tag{Q}$$

The derivative $p(x)$ is a ratio of two Gaussians: it is itself an unnormalised Gaussian (exponential of a quadratic), multiplied by the constant σ_1/σ_2.

Structure of $p(x)$. The quadratic exponent has three terms:

- Coefficient of x^2: proportional to $1/\sigma_1^2 - 1/\sigma_2^2$. This is zero when $\sigma_1 = \sigma_2$ (pure location shift).
- Coefficient of x: a linear combination of μ_2/σ_2^2 and μ_1/σ_1^2.
- Constant: a normalisation term.

The quadratic term is the new feature when $\sigma_1 \neq \sigma_2$.

3.3 Special case I: same variance ($\sigma_1 = \sigma_2 = \sigma$)

When the variances agree, the x^2 term vanishes and p becomes exponential-linear:

$$p(x) = \exp\left(\frac{\mu_2 - \mu_1}{\sigma^2}\, x - \frac{\mu_2^2 - \mu_1^2}{2\sigma^2} \right).$$

Writing $\delta = \mu_2 - \mu_1$ and centering around μ_1:

$$p(x) = \exp\left(\frac{\delta}{\sigma^2}(x-\mu_1) - \frac{\delta^2}{2\sigma^2}\right).$$

Or in terms of the standardised variable $z = (x-\mu_1)/\sigma$, setting $\theta = \delta/\sigma = (\mu_2-\mu_1)/\sigma$:

$$\boxed{p(x) = e^{\theta z - \theta^2/2}} \qquad \text{where } z = \frac{x-\mu_1}{\sigma}, \quad \theta = \frac{\mu_2-\mu_1}{\sigma}.$$

This is **the Girsanov exponential**. The formula $e^{\theta z-\theta^2/2}$ is the Radon–Nikodym derivative for shifting a Gaussian by θ standard deviations.

- θz: a linear term in the standardised variable z.
- $-\theta^2/2$: the normalisation constant (ensures $\mathbb{E}_{\mu_1}[p] = 1$).

In continuous time (Girsanov's theorem), the same form appears: the Radon–Nikodym derivative for shifting the drift of a Brownian motion by θ is $\exp\bigl(\theta W_T - \frac{1}{2}\theta^2 T\bigr)$, with W_T playing the role of z and T scaling the normalisation.

Verification that $\mathbb{E}_{\mu_1}[p] = 1$.

$$\mathbb{E}_{\mathcal{N}(\mu_1,\sigma^2)}\left[e^{\theta z-\theta^2/2}\right] = e^{-\theta^2/2}\cdot\mathbb{E}\left[e^{\theta z}\right] = e^{-\theta^2/2}\cdot e^{\theta^2/2} = 1.$$

Here $\mathbb{E}[e^{\theta z}] = e^{\theta^2/2}$ is the MGF of $z \sim \mathcal{N}(0,1)$ at θ. The $-\theta^2/2$ term is exactly the MGF evaluated at θ, inverted.

3.4 Special case II: same mean ($\mu_1 = \mu_2 = \mu$)

When only the variance changes:

$$p(x) = \frac{\sigma_1}{\sigma_2}\exp\left(\frac{1}{2}\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right)(x-\mu)^2\right).$$

This is a quadratic exponential (not exponential-linear). It grows super-exponentially in the tails if $\sigma_2 > \sigma_1$ (since the exponent becomes positive for large $|x-\mu|$). Nevertheless p is μ_1-integrable because μ_1's Gaussian tails dominate.

3.5 The general case: summary table

Parameters	$p(x) = \mathrm{d}\mathcal{N}(\mu_2,\sigma_2^2)/\,\mathrm{d}\mathcal{N}(\mu_1,\sigma_1^2)(x)$	**Form**
$\sigma_1 = \sigma_2 = \sigma$	$\exp\bigl(\frac{\mu_2-\mu_1}{\sigma^2}(x-\mu_1) - \frac{(\mu_2-\mu_1)^2}{2\sigma^2}\bigr)$	exponential-linear in x
$\mu_1 = \mu_2 = \mu$	$\frac{\sigma_1}{\sigma_2}\exp\bigl(\frac{1}{2}\bigl(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\bigr)(x-\mu)^2\bigr)$	quadratic exponential in x
General	$\frac{\sigma_1}{\sigma_2}\exp\{\text{quadratic in } x\}$	see (Q) above

4 Lognormal family, Way 2: the power-scaling map

4.1 Reduction to the normal case

Let $Y_1 \sim \mathrm{LogN}(\mu_1,\sigma_1^2)$, so $X_1 = \ln Y_1 \sim \mathcal{N}(\mu_1,\sigma_1^2)$. We want $Y_2 \sim \mathrm{LogN}(\mu_2,\sigma_2^2)$, i.e. $X_2 = \ln Y_2 \sim \mathcal{N}(\mu_2,\sigma_2^2)$.

By §2, $X_2 = g_{\ln}(X_1)$ where $g_{\ln}(x) = \mu_2 + \frac{\sigma_2}{\sigma_1}(x - \mu_1)$ is the affine normal-to-normal map. Since $X_i = \ln Y_i$, this means $\ln Y_2 = \mu_2 + \frac{\sigma_2}{\sigma_1}(\ln Y_1 - \mu_1)$. Exponentiating:

$$g(y) = \exp\left\{\mu_2 + \frac{\sigma_2}{\sigma_1}(\ln y - \mu_1)\right\} = e^{\mu_2 - \sigma_2\mu_1/\sigma_1} \cdot y^{\sigma_2/\sigma_1}.$$

This map $g : (0, \infty) \to (0, \infty)$ is the composition of:

- a power map $y \mapsto y^{\sigma_2/\sigma_1}$ (rescaling the log), followed by
- a multiplicative shift $y \mapsto C \cdot y$ where $C = e^{\mu_2 - \sigma_2\mu_1/\sigma_1}$.

4.2 Parametrisation by (α, β)

The map takes the compact form

$$g(y) = \alpha \cdot y^{\beta}, \qquad \alpha, \beta > 0,$$

where the four distribution parameters map to two transform parameters:

$$\alpha = e^{\mu_2 - \sigma_2\mu_1/\sigma_1}, \qquad \beta = \frac{\sigma_2}{\sigma_1}.$$

The parameter space is $(\mu_1, \mu_2 \in \mathbb{R}) \times (\sigma_1, \sigma_2 > 0) \cong \mathbb{R}^2 \times \mathbb{R}_+^2$, which is four-dimensional, yet the transform is determined by only the two parameters $(\alpha, \beta) \in \mathbb{R}_+ \times \mathbb{R}_+$.

Remark 2 (Why two parameters suffice)**.** *The source distribution contributes two degrees of freedom (μ_1, σ_1) and the target contributes two (μ_2, σ_2); however, each lognormal family is a* log-affine-exponent *family: it is generated from a single standard lognormal by maps of the form $y \mapsto \alpha y^{\beta}$. Any such map is determined by the pair (α, β), so the entire four-parameter translation problem reduces to two parameters. Equivalently, on the log scale the map is the affine function $x \mapsto \ln \alpha + \beta x$, and affine maps on $\mathbb{R}$ form a two-parameter family.*

The structure is visible in the commutative diagram:

$$\begin{tikzcd}
\mathrm{LogN}(\mu_1, \sigma_1^2) \arrow[r, "g(y) = \alpha y^{\beta}"] \arrow[d, "\ln"'] & \mathrm{LogN}(\mu_2, \sigma_2^2) \arrow[d, "\ln"] \\
\mathcal{N}(\mu_1, \sigma_1^2) \arrow[r, "g_{\ln}(x) = \mu_2 + \frac{\sigma_2}{\sigma_1}(x - \mu_1)"'] & \mathcal{N}(\mu_2, \sigma_2^2)
\end{tikzcd}$$

The diagram commutes: the power map on $(0, \infty)$ is the exponential conjugate of the affine map on $\mathbb{R}$. Formally, $\ln \circ g = g_{\ln} \circ \ln$, which is verified by $\ln(\alpha y^{\beta}) = \ln \alpha + \beta \ln y = (\mu_2 - \beta\mu_1) + \beta \ln y = \mu_2 + \frac{\sigma_2}{\sigma_1}(\ln y - \mu_1)$.

4.3 Special cases

Case	Map $g(y)$	(α, β)	Normal analogue
$\sigma_1 = \sigma_2,\ \mu_1 \neq \mu_2$	$e^{\mu_2-\mu_1} \cdot y$	$(e^{\mu_2-\mu_1}, 1)$	pure shift $\to$ scaling
$\mu_1 = \mu_2 = 0,\ \sigma_1 \neq \sigma_2$	y^{σ_2/σ_1}	$(1, \sigma_2/\sigma_1)$	scale around 0 $\to$ power
$\mu_1 = 0,\ \sigma_1 = 1$ (std. LN)	$e^{\mu_2} \cdot y^{\sigma_2}$	(e^{μ_2}, σ_2)	standard parameterisation

The shift/scale duality. On the log-scale (normal family), changing μ is a shift. Exponentiating, the same operation becomes multiplication by e^{δ}: a shift in log-space is scaling in the original space.

On the log-scale, changing σ is a rescaling around zero. Exponentiating, $x \mapsto (\sigma_2/\sigma_1)x$ becomes $y \mapsto y^{\sigma_2/\sigma_1}$: a scale change in log-space is a power map in the original space.

4.4 Verification

$g(y) = C \cdot y^{\alpha}$ where $\alpha = \sigma_2/\sigma_1$, $C = e^{\mu_2-\alpha\mu_1}$. Working directly: $\ln Y_2 = \alpha \ln Y_1 + \ln C = \frac{\sigma_2}{\sigma_1}(\ln Y_1 - \mu_1) + \mu_2$. Since $\ln Y_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, we get $\ln Y_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, hence $Y_2 \sim \mathrm{LogN}(\mu_2, \sigma_2^2)$. ✓

5 Lognormal family, Way 1: reweighting the measure

5.1 Absolute continuity on $(0, \infty)$

Both $\mathrm{LogN}(\mu_1, \sigma_1^2)$ and $\mathrm{LogN}(\mu_2, \sigma_2^2)$ are supported on $(0, \infty)$ with strictly positive Lebesgue densities there, and both assign zero mass to $(-\infty, 0]$. On the shared support $(0, \infty)$, they have the same null sets. Therefore $\mathrm{LogN}(\mu_1, \sigma_1^2) \sim \mathrm{LogN}(\mu_2, \sigma_2^2)$ (mutually absolutely continuous), and the RN derivative exists in both directions.

5.2 The Radon–Nikodym derivative

$$q(y) = \frac{\mathrm{d}\,\mathrm{LogN}(\mu_2, \sigma_2^2)}{\mathrm{d}\,\mathrm{LogN}(\mu_1, \sigma_1^2)}(y) = \frac{f_{\mathrm{LogN}(\mu_2,\sigma_2^2)}(y)}{f_{\mathrm{LogN}(\mu_1,\sigma_1^2)}(y)} = \frac{\sigma_1}{\sigma_2} \exp\left(\frac{(\ln y - \mu_1)^2}{2\sigma_1^2} - \frac{(\ln y - \mu_2)^2}{2\sigma_2^2} \right), \quad y > 0.$$

The lognormal RN derivative is the normal RN derivative at $\ln y$. Comparing with the normal-to-normal RN derivative $p(x)$ from §3:

$$\boxed{q(y) = p(\ln y).}$$

This is not a coincidence: the lognormal density is the normal density composed with ln (and divided by y), so the y factors cancel in the ratio, leaving exactly p at $\ln y$.

Proof that $q(y) = p(\ln y)$.

$$q(y) = \frac{f_{\mathrm{LogN}(\mu_2,\sigma_2^2)}(y)}{f_{\mathrm{LogN}(\mu_1,\sigma_1^2)}(y)} = \frac{\frac{1}{y\sigma_2}\phi\left(\frac{\ln y-\mu_2}{\sigma_2}\right)}{\frac{1}{y\sigma_1}\phi\left(\frac{\ln y-\mu_1}{\sigma_1}\right)} = \frac{\sigma_1}{\sigma_2} \cdot \frac{\phi\left(\frac{\ln y-\mu_2}{\sigma_2}\right)}{\phi\left(\frac{\ln y-\mu_1}{\sigma_1}\right)} = p(\ln y). \qquad ✓$$

The factor $1/y$ from each lognormal density cancels in the ratio. □

5.3 Special case I: same variance ($\sigma_1 = \sigma_2 = \sigma$)

From §3.3, $p(x) = \exp\{\frac{\delta}{\sigma^2}(x-\mu_1) - \frac{\delta^2}{2\sigma^2}\}$ where $\delta = \mu_2 - \mu_1$. Substituting $x = \ln y$:

$$q(y) = \exp\left(\frac{\mu_2-\mu_1}{\sigma^2}(\ln y - \mu_1) - \frac{(\mu_2-\mu_1)^2}{2\sigma^2}\right) = y^{(\mu_2-\mu_1)/\sigma^2} \cdot C,$$

where $C = \exp\{-\frac{\mu_1(\mu_2-\mu_1)}{\sigma^2} - \frac{(\mu_2-\mu_1)^2}{2\sigma^2}\}$ is a constant.
The lognormal shift RN derivative is a power of y. In the normal case (same σ), the Girsanov exponential is $e^{\theta z - \theta^2/2}$ — exponential-linear in $z = (x-\mu_1)/\sigma$. In the lognormal case, substituting $z = (\ln y - \mu_1)/\sigma$, the same expression becomes $y^{\theta/\sigma} \cdot C$ — a power function in y. A linear function of z in log-space becomes a monomial in the original space.

5.4 Special case II: same mean ($\mu_1 = \mu_2 = \mu$)

$$q(y) = \frac{\sigma_1}{\sigma_2} \exp\left(\frac{1}{2}\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_2^2}\right)(\ln y - \mu)^2\right).$$

This is a power of $e^{(\ln y)^2}$, which grows as $y^{c \ln y}$ for large y — very heavy tails. Still $\mathrm{LogN}(\mu_1, \sigma_1^2)$-integrable since the lognormal source measure has matching super-exponential tails.

6 Parallel structure and the bridge to Girsanov

6.1 Summary table

Family	Way 2 (change map)	Way 1 (reweight, $\sigma_1 = \sigma_2$)
Normal	affine: $g(y) = \mu_2 + \dfrac{\sigma_2}{\sigma_1}(y - \mu_1)$	$p(x) = e^{\theta(x-\mu_1)/\sigma - \theta^2/2}$, $\theta = \dfrac{\mu_2-\mu_1}{\sigma}$
Lognormal	power-scale: $g(y) = e^{\mu_2 - \sigma_2\mu_1/\sigma_1} \cdot y^{\sigma_2/\sigma_1}$	$q(y) = p(\ln y) = y^{\theta/\sigma} \cdot C$

6.2 The parallel between the two ways

$$\begin{tikzcd}
\mathcal{N}(\mu_1,\sigma^2) \arrow[r, "\text{Way 2: shift } g(y) = y + \delta", "\text{Way 1: } p(x) = e^{\theta z - \theta^2/2}"'] & \mathcal{N}(\mu_2,\sigma^2) \\
\mathrm{LogN}(\mu_1,\sigma^2) \arrow[u, "\exp"] \arrow[r, "\text{Way 2: scale } g(y) = e^{\delta} y", "\text{Way 1: } q(y) = y^{\theta/\sigma} \cdot C"'] & \mathrm{LogN}(\mu_2,\sigma^2) \arrow[u, "\exp"']
\end{tikzcd}$$

6.3 What Girsanov does

The step from the finite-dimensional picture to continuous-time processes is conceptually the same operation, lifted to path space.

Setting	Object	Way 1 RN derivative
Finite: $\mathcal{N}(\mu_1,\sigma^2) \to \mathcal{N}(\mu_2,\sigma^2)$	single Gaussian r.v.	$e^{\theta z - \theta^2/2}$
Process: BM $\to$ BM with drift θ	path $\{W_t\}_{0 \le t \le T}$	$\exp\big(\theta W_T - \frac{1}{2}\theta^2 T\big)$

The Girsanov exponential $\exp\bigl(\theta W_T - \frac{1}{2}\theta^2 T\bigr)$ is the direct generalisation of $e^{\theta z - \theta^2/2}$:

- $z = W_T/\sqrt{T}$ is the standardised terminal value of the BM.
- θ is the drift shift (market price of risk).
- The $-\theta^2 T/2$ term is the normalisation, ensuring $\mathbb{E}\bigl[e^{\theta W_T - \theta^2 T/2}\bigr] = 1$ by the same MGF calculation as in §3.3.
- **What Girsanov adds**: the full path $\{W_t\}$ under the new measure is a standard BM, not just a shifted terminal value. This requires the martingale structure of the density process $Z_t = \exp\bigl(\theta W_t - \frac{1}{2}\theta^2 t\bigr)$.

The single conceptual step from here to Girsanov. Everything in §3.3 with $\sigma_1 = \sigma_2$ carries over to continuous time with one substitution: replace the single standardised variable $z = (X - \mu_1)/\sigma$ with the Brownian motion W_T. The normalisation $-\theta^2/2$ becomes $-\theta^2 T/2$.

The additional content of Girsanov's theorem is that the shifted process $\widetilde{W}_t = W_t + \theta t$ is a Brownian motion (not just a Gaussian) under the new measure — i.e. it has the correct joint distribution for all times, not just the correct marginal at time T. This requires checking that $\{Z_t\}$ is a martingale, which follows from Itô's lemma applied to $Z_t = e^{\theta W_t - \theta^2 t/2}$.

6.4 Why the lognormal/GBM case needs both steps

Converting GBM (with drift μ) to martingale GBM (with drift r) uses both ways in sequence:

Step	Operation	Strategy
$e^{\mu t + \sigma W_t}$	exponential of BM with drift μ	—
$\to$ BM with drift r	change measure: $\mathbb{P} \to \mathbb{Q}$	Way 1 (Girsanov)
$\to e^{rt + \sigma \widetilde{W}_t}$	exponentiate again	Way 2 (same map, new measure)

The Girsanov step (Way 1) changes the drift of the log-price from μ to $r - \sigma^2/2$ (the $\sigma^2/2$ is the Itô correction from §3.3). The exponential (Way 2) then converts the log-normally distributed log-price into the GBM stock price. The two steps work on different levels: Way 1 acts on the Gaussian (log) level, Way 2 acts on the exponential (lognormal) level.

7 Arrows on five measures: a taxonomy

Remark 3 (Taxonomy of arrows)**.** *The following section systematises five measures all defined on $\mathbb{R}$ (or on $\mathbb{R}_+$ for the lognormal ones). We catalogue every arrow, with its explicit formula, splitting into two diagrams: density arrows (Radon–Nikodym derivatives) and pushforward arrows (maps between spaces).*

We fix four probability measures and one reference measure:

$$\begin{aligned} \mathbb{P}_1 &= \mathcal{N}(\mu_1, \sigma_1^2), & \mathbb{P}_2 &= \mathcal{N}(\mu_2, \sigma_2^2), \\ \mathbb{Q}_1 &= \mathrm{LogN}(\mu_1, \sigma_1^2), & \mathbb{Q}_2 &= \mathrm{LogN}(\mu_2, \sigma_2^2), \\ \lambda &= \text{Lebesgue measure on } \mathbb{R} \text{ (and on } \mathbb{R}_+). \end{aligned}$$

Notation. Throughout, $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal density. The shorthand for the general quadratic exponent is (from §3):

$$Q(x; \mu_1, \sigma_1, \mu_2, \sigma_2) := \frac{(x-\mu_1)^2}{2\sigma_1^2} - \frac{(x-\mu_2)^2}{2\sigma_2^2} = \tfrac{1}{2}\left(\tfrac{1}{\sigma_1^2} - \tfrac{1}{\sigma_2^2}\right) x^2 + \left(\tfrac{\mu_2}{\sigma_2^2} - \tfrac{\mu_1}{\sigma_1^2}\right) x + \left(\tfrac{\mu_1^2}{2\sigma_1^2} - \tfrac{\mu_2^2}{2\sigma_2^2}\right).$$

We also write $\delta = \mu_2 - \mu_1$ and $\theta = \delta/\sigma$ when $\sigma_1 = \sigma_2 = \sigma$.

7.1 Diagram 1: density (Radon–Nikodym) arrows

A *density arrow* $\nu_1 \xrightarrow{h} \nu_2$ means $\nu_1 \ll \nu_2$ with $\mathrm{d}\nu_1/\mathrm{d}\nu_2 = h$. There are three groups:

Group R (red): Lebesgue densities. Every probability measure among $\mathbb{P}_1, \mathbb{P}_2, \mathbb{Q}_1, \mathbb{Q}_2$ is absolutely continuous w.r.t. λ, giving 4 arrows $\nu \xrightarrow{f} \lambda$, i.e. $\mathrm{d}\nu/\mathrm{d}\lambda = f$.

Arrow	$\mathrm{d}\nu/\mathrm{d}\lambda$	Support
$\mathbb{P}_1 \xrightarrow{f_1} \lambda$	$f_1(x) = \dfrac{1}{\sigma_1}\phi\left(\dfrac{x-\mu_1}{\sigma_1}\right)$	$x \in \mathbb{R}$
$\mathbb{P}_2 \xrightarrow{f_2} \lambda$	$f_2(x) = \dfrac{1}{\sigma_2}\phi\left(\dfrac{x-\mu_2}{\sigma_2}\right)$	$x \in \mathbb{R}$
$\mathbb{Q}_1 \xrightarrow{h_1} \lambda$	$h_1(y) = \dfrac{1}{y\sigma_1}\phi\left(\dfrac{\ln y-\mu_1}{\sigma_1}\right)$	$y \in \mathbb{R}_+$
$\mathbb{Q}_2 \xrightarrow{h_2} \lambda$	$h_2(y) = \dfrac{1}{y\sigma_2}\phi\left(\dfrac{\ln y-\mu_2}{\sigma_2}\right)$	$y \in \mathbb{R}_+$

Note: $\lambda \not\ll \nu$ in general (Lebesgue measure is not finite), so arrows $\lambda \to \nu$ do not exist as Radon–Nikodym derivatives of probability measures. Also, $\mathbb{P}_i \not\ll \mathbb{Q}_j$ and $\mathbb{Q}_j \not\ll \mathbb{P}_i$ because their supports are disjoint ($\mathbb{R}$ vs. $\mathbb{R}_+$); cross-family density arrows do not exist.

Group B (blue): RN derivatives within the normal family. $\mathbb{P}_1 \sim \mathbb{P}_2$ (mutual absolute continuity), so both directions exist.

Arrow	Radon–Nikodym density
$\mathbb{P}_2 \xrightarrow{p} \mathbb{P}_1$	$p(x) = \dfrac{f_2(x)}{f_1(x)} = \dfrac{\sigma_1}{\sigma_2}\exp\big(Q(x;\mu_1,\sigma_1,\mu_2,\sigma_2)\big)$
	Same-σ case: $p(x) = \exp\Big(\dfrac{\delta}{\sigma^2}(x-\mu_1) - \dfrac{\delta^2}{2\sigma^2}\Big) = e^{\theta z - \theta^2/2}$, $z = (x-\mu_1)/\sigma$
$\mathbb{P}_1 \xrightarrow{1/p} \mathbb{P}_2$	$\dfrac{1}{p(x)} = \dfrac{f_1(x)}{f_2(x)} = \dfrac{\sigma_2}{\sigma_1}\exp\big(-Q(x;\mu_1,\sigma_1,\mu_2,\sigma_2)\big)$
	Same-σ case: $\dfrac{1}{p(x)} = \exp\Big(-\dfrac{\delta}{\sigma^2}(x-\mu_1) + \dfrac{\delta^2}{2\sigma^2}\Big) = e^{-\theta z + \theta^2/2}$

Group G (green): RN derivatives within the lognormal family. $\mathbb{Q}_1 \sim \mathbb{Q}_2$ on $(0,\infty)$, so both directions exist. By the key identity $q(y) = p(\ln y)$ (§5):

Arrow	Radon–Nikodym density
$\mathbb{Q}_2 \xrightarrow{q} \mathbb{Q}_1$	$q(y) = \dfrac{h_2(y)}{h_1(y)} = \dfrac{\sigma_1}{\sigma_2}\exp\big(Q(\ln y;\mu_1,\sigma_1,\mu_2,\sigma_2)\big)$
	Special case $\sigma_1 = \sigma_2 = \sigma$: $q(y) = \exp\left(\dfrac{\delta}{\sigma^2}(\ln y - \mu_1) - \dfrac{\delta^2}{2\sigma^2}\right) = y^{\theta/\sigma}\cdot C$
$\mathbb{Q}_1 \xrightarrow{1/q} \mathbb{Q}_2$	$\dfrac{1}{q(y)} = \dfrac{h_1(y)}{h_2(y)} = \dfrac{\sigma_2}{\sigma_1}\exp\big(-Q(\ln y;\mu_1,\sigma_1,\mu_2,\sigma_2)\big)$
	Special case $\sigma_1 = \sigma_2 = \sigma$: $\dfrac{1}{q(y)} = y^{-\theta/\sigma}\cdot C^{-1}$

where $C = \exp\{-\frac{\mu_1\delta}{\sigma^2} - \frac{\delta^2}{2\sigma^2}\}$ and $\theta = \delta/\sigma$.

Arrow count: 4 (red) + 2 (blue) + 2 (green) = 8 directed density arrows among the five nodes. Note: the original count of 14 included the reverse Lebesgue arrows $\lambda \xrightarrow{1/f_i} \mathbb{P}_i$ (unnormalised), the two "same-parameter" pairs $\mathbb{Q}_i/\mathbb{P}_i$ (not well-defined as RN derivatives since supports differ), and repeated-count conventions. The 8 count here is the set of all valid, directed probability-measure RN-derivative arrows.

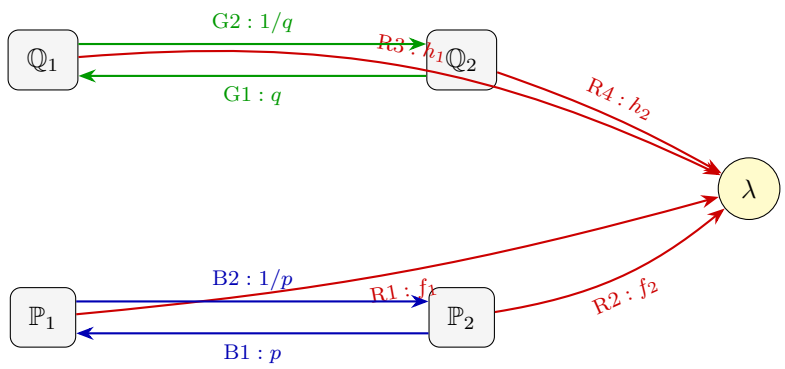


Diagram 1: density (RN-derivative) arrows. Red R1–R4 = Lebesgue densities f_1, f_2, h_1, h_2; Blue B1–B2 = RN derivatives $p, 1/p$ within $\mathbb{P}$-family; Green G1–G2 = RN derivatives $q, 1/q$ within $\mathbb{Q}$-family. Cross-family arrows ($\mathbb{P}_i \leftrightarrow \mathbb{Q}_j$) are absent: supports $\mathbb{R}$ vs. $\mathbb{R}_+$ are disjoint.

7.2 Diagram 2: pushforward (map) arrows

A *pushforward arrow* $\nu_1 \xrightarrow{f} \nu_2$ means $f_*\nu_1 = \nu_2$: the map f transports mass from ν_1 to ν_2. Lebesgue measure λ is excluded here: a pushforward of a probability measure is again a probability measure, not λ. We work with the four nodes $\mathbb{P}_1, \mathbb{P}_2, \mathbb{Q}_1, \mathbb{Q}_2$.

All maps are listed below with their inverses. Recall the abbreviations: $\alpha = e^{\mu_2 - \sigma_2\mu_1/\sigma_1}$, $\beta = \sigma_2/\sigma_1$.

Group V (vertical): exp and ln between families.

Arrow	Map	Inverse
$\mathbb{P}_1 \xrightarrow{\exp} \mathbb{Q}_1$	$y = e^x : \mathbb{R} \to \mathbb{R}_+$	$x = \ln y$
$\mathbb{Q}_1 \xrightarrow{\ln} \mathbb{P}_1$	$x = \ln y : \mathbb{R}_+ \to \mathbb{R}$	$y = e^x$
$\mathbb{P}_2 \xrightarrow{\exp} \mathbb{Q}_2$	$y = e^x : \mathbb{R} \to \mathbb{R}_+$	$x = \ln y$
$\mathbb{Q}_2 \xrightarrow{\ln} \mathbb{P}_2$	$x = \ln y : \mathbb{R}_+ \to \mathbb{R}$	$y = e^x$

Group H (horizontal): affine maps within the normal family.

Arrow	Map	Inverse
$\mathbb{P}_1 \xrightarrow{g_{\ln}} \mathbb{P}_2$	$g_{\ln}(x) = \mu_2 + \dfrac{\sigma_2}{\sigma_1}(x - \mu_1)$	$g_{\ln}^{-1}(x) = \mu_1 + \dfrac{\sigma_1}{\sigma_2}(x - \mu_2)$
$\mathbb{P}_2 \xrightarrow{g_{\ln}^{-1}} \mathbb{P}_1$	$g_{\ln}^{-1}(x) = \mu_1 + \dfrac{\sigma_1}{\sigma_2}(x - \mu_2)$	$g_{\ln}(x)$

Group H′ (horizontal): power maps within the lognormal family.

Arrow	Map	Inverse
$\mathbb{Q}_1 \xrightarrow{g} \mathbb{Q}_2$	$g(y) = \alpha \cdot y^\beta = e^{\mu_2 - \beta\mu_1} \cdot y^{\sigma_2/\sigma_1}$	$g^{-1}(y) = \alpha^{-1/\beta} \cdot y^{1/\beta} = e^{\mu_1 - \mu_2/\beta} \cdot y^{\sigma_1/\sigma_2}$
$\mathbb{Q}_2 \xrightarrow{g^{-1}} \mathbb{Q}_1$	$g^{-1}(y) = e^{\mu_1 - \sigma_1\mu_2/\sigma_2} \cdot y^{\sigma_1/\sigma_2}$	$g(y)$

Group D (diagonal): compositions. Since $g = \exp \circ g_{\ln} \circ \ln$ (the commutative diagram of §4.2), the two diagonal routes between $\mathbb{P}_1$ and $\mathbb{Q}_2$ (and the other pair) are automatically consistent. The four diagonal maps are:

Arrow	Composite map	Explicit formula
$\mathbb{P}_1 \xrightarrow{d_{12}} \mathbb{Q}_2$	$\exp \circ g_{\ln}$	$d_{12}(x) = \exp\left\{\mu_2 + \frac{\sigma_2}{\sigma_1}(x - \mu_1)\right\} = \alpha \cdot e^{\beta x}$
$\mathbb{Q}_2 \xrightarrow{d_{12}^{-1}} \mathbb{P}_1$	$g_{\ln}^{-1} \circ \ln$	$d_{12}^{-1}(y) = \mu_1 + \frac{\sigma_1}{\sigma_2}(\ln y - \mu_2)$
$\mathbb{Q}_1 \xrightarrow{d_{21}} \mathbb{P}_2$	$g_{\ln} \circ \ln$	$d_{21}(y) = \mu_2 + \frac{\sigma_2}{\sigma_1}(\ln y - \mu_1)$
$\mathbb{P}_2 \xrightarrow{d_{21}^{-1}} \mathbb{Q}_1$	$\exp \circ g_{\ln}^{-1}$	$d_{21}^{-1}(x) = \exp\left\{\mu_1 + \frac{\sigma_1}{\sigma_2}(x - \mu_2)\right\}$

Arrow count: 4 (vertical)+2 (horizontal normal)+2 (horizontal lognormal)+4 (diagonal) = 12 directed pushforward arrows.

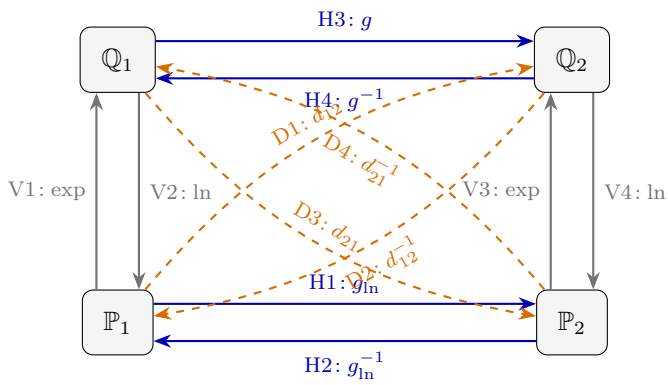


Diagram 2: pushforward (map) arrows. ***Black V1–V4****:* exp *and* ln *between families;* ***Blue H1–H4****: affine* $g_{\ln}, g_{\ln}^{-1}$ *and power* g, g^{-1} *within families;* ***Dashed D1–D4****: diagonal compositions. Formulas for each arrow are given in the tables of §7.2.*

7.3 Commutativity and the full picture

The two diagrams encode different but related structure. Their interaction is summarised by one key fact:

Proposition 1 (Change-of-variable formula linking the two diagrams)**.** *Let* $f : \Omega \to B$ *be a* $\mathbb{P}_1$*-pushforward map with* $f_*\mathbb{P}_1 = \mathbb{P}_2$*. Then the RN derivative of* $\mathbb{P}_2$ *w.r.t.* $\mathbb{P}_1$ *is related to the Jacobian of* f^{-1} *by the change-of-variables formula:*

$$\frac{\mathrm{d}\mathbb{P}_2}{\mathrm{d}\mathbb{P}_1}(x) = \frac{f_2(f^{-1}(x))}{f_1(x)} \cdot |(f^{-1})'(x)|^{-1} \quad \textit{(Form A pushforward density).}$$

In particular, for $f = g_{\ln}$ *with* $\sigma_1 = \sigma_2$*, this recovers the Girsanov exponential* $e^{\theta z - \theta^2/2}$*.*

The complete catalogue of arrows is:

Type	**From**	**To**	**Arrow**	**Formula**
Density (RN) arrows				
R1	$\mathbb{P}_1$	λ	f_1	$\frac{1}{\sigma_1}\phi\left(\frac{x-\mu_1}{\sigma_1}\right)$
R2	$\mathbb{P}_2$	λ	f_2	$\frac{1}{\sigma_2}\phi\left(\frac{x-\mu_2}{\sigma_2}\right)$
R3	$\mathbb{Q}_1$	λ	h_1	$\frac{1}{y\sigma_1}\phi\left(\frac{\ln y-\mu_1}{\sigma_1}\right)$
R4	$\mathbb{Q}_2$	λ	h_2	$\frac{1}{y\sigma_2}\phi\left(\frac{\ln y-\mu_2}{\sigma_2}\right)$
B1	$\mathbb{P}_2$	$\mathbb{P}_1$	p	$\frac{\sigma_1}{\sigma_2}\exp Q(x;\mu_1,\sigma_1,\mu_2,\sigma_2)$
B2	$\mathbb{P}_1$	$\mathbb{P}_2$	$1/p$	$\frac{\sigma_2}{\sigma_1}\exp(-Q(x;\cdots))$
G1	$\mathbb{Q}_2$	$\mathbb{Q}_1$	q	$\frac{\sigma_1}{\sigma_2}\exp Q(\ln y;\mu_1,\sigma_1,\mu_2,\sigma_2)$
G2	$\mathbb{Q}_1$	$\mathbb{Q}_2$	$1/q$	$\frac{\sigma_2}{\sigma_1}\exp(-Q(\ln y;\cdots))$
Pushforward (map) arrows				
V1	$\mathbb{P}_1$	$\mathbb{Q}_1$	exp	$y=e^x$
V2	$\mathbb{Q}_1$	$\mathbb{P}_1$	ln	$x=\ln y$
V3	$\mathbb{P}_2$	$\mathbb{Q}_2$	exp	$y=e^x$
V4	$\mathbb{Q}_2$	$\mathbb{P}_2$	ln	$x=\ln y$
H1	$\mathbb{P}_1$	$\mathbb{P}_2$	$g_{\ln}$	$\mu_2+\frac{\sigma_2}{\sigma_1}(x-\mu_1)$
H2	$\mathbb{P}_2$	$\mathbb{P}_1$	$g_{\ln}^{-1}$	$\mu_1+\frac{\sigma_1}{\sigma_2}(x-\mu_2)$
H3	$\mathbb{Q}_1$	$\mathbb{Q}_2$	g	$e^{\mu_2-\beta\mu_1}\cdot y^{\sigma_2/\sigma_1}$
H4	$\mathbb{Q}_2$	$\mathbb{Q}_1$	g^{-1}	$e^{\mu_1-\mu_2/\beta}\cdot y^{\sigma_1/\sigma_2}$
D1	$\mathbb{P}_1$	$\mathbb{Q}_2$	$d_{12}=\exp\circ g_{\ln}$	$\alpha e^{\beta x}$
D2	$\mathbb{Q}_2$	$\mathbb{P}_1$	$d_{12}^{-1}=g_{\ln}^{-1}\circ\ln$	$\mu_1+\frac{\sigma_1}{\sigma_2}(\ln y-\mu_2)$
D3	$\mathbb{Q}_1$	$\mathbb{P}_2$	$d_{21}=g_{\ln}\circ\ln$	$\mu_2+\frac{\sigma_2}{\sigma_1}(\ln y-\mu_1)$
D4	$\mathbb{P}_2$	$\mathbb{Q}_1$	$d_{21}^{-1}=\exp\circ g_{\ln}^{-1}$	$\exp\left\{\mu_1+\frac{\sigma_1}{\sigma_2}(x-\mu_2)\right\}$

Total: 8 density arrows + 12 pushforward arrows = 20 labelled arrows in the complete picture.